FutureWorlds: Branch-Specific Memory for Long-Horizon Robotic World Models

Anonymous CVPR submission

Abstract

Robotic world models predict future scenes conditioned on actions, providing a foundation for understanding action outcomes. Reinforcement learning improves these models through trajectory rewards, but effective learning requires two conditions: informative comparisons among candidate futures and consistency between policy scoring and generation. Reinforcement learning for autoregressive world models faces obstacles in both respects: conventional beam search suffers from candidate collapse that shrinks within-group reward differences and weakens the learning signal from relative advantages, whereas independent sampling scatters into low-likelihood trajectories; meanwhile, history mismatch makes policy scoring depart from the conditions used for generation. We introduce FUTUREWORLDS*, a reinforcement learning framework built on a multimodal discrete autoregressive world model. The framework uses diverse beam search to construct candidate futures that balance model confidence and diversity, and maintains bounded memory for each branch that inherits its parent's history and evolves during generation, ensuring that generation and policy scoring use the same history. As its core training mechanism,* MEMSPO *(Memory-Conditioned Search-Guided Policy Optimization) converts video trajectory rewards into group-relative advantages to optimize the world model. Experiments across multiple robotic manipulation datasets show that, under matched initialization and training budgets, standard GRPO offers limited gains over supervised fine-tuning, whereas* MEMSPO *delivers consistent improvements, reducing long-horizon prediction LPIPS by up to 9.75% and 9.98% relative to supervised fine-tuning and GRPO, respectively. Ablations, optical-flow analysis, and qualitative comparisons further show that these gains manifest in more accurate motion prediction and more consistent object states. Our code, checkpoints, and project page are available at this anonymous URL.*

1. Introduction

Robotic world models predict how scenes evolve in response to actions, providing a foundation for action-outcome prediction [10], visual planning [9, 14], behavior learning [15], and policy evaluation [12, 40]. Given the same observations, action sequence, and instruction, a model can generate multiple predictions of the future. These candidate futures represent different hypotheses about scene evolution and vary in robot motion, object states, and prediction quality. Their differences offer a learning opportunity: by constructing distinct candidates and comparing them against real observations, a world model can learn from its own predictions.

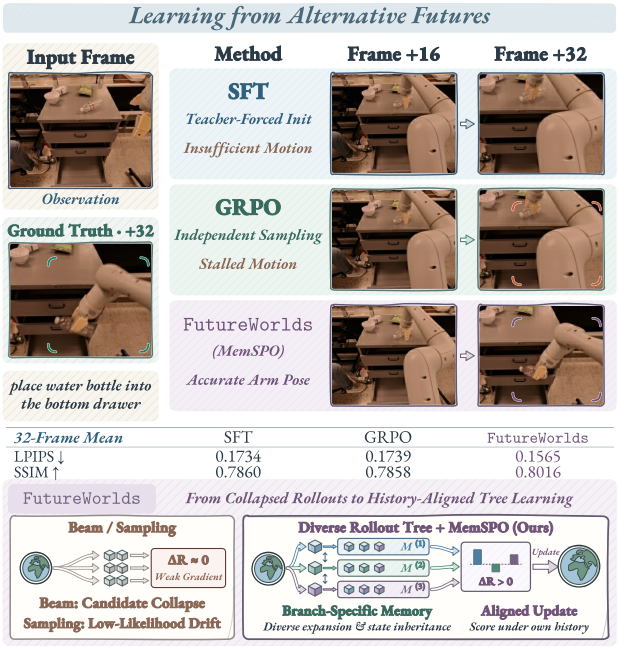


Figure 1. ***Learning from alternative futures.*** Representative RT-1 predictions from a shared SFT initialization. Metrics average 32 frames over 128 cases. GRPO and MEMSPO use matched 200-update checkpoints and identical Beam4 decoding. Bottom: diverse futures and history-aligned learning.

Most autoregressive world models learn from real trajectories through supervised training [33]. During training, they condition on ground-truth histories; at inference, they rely on self-generated histories, allowing errors to accumulate throughout a rollout [4, 26]. Token-level likelihood also does not directly optimize complete generated sequences [24, 25]. Reinforcement learning (RL) connects generated outputs

to learning objectives through sequence-level rewards [5]. In world modeling, RLVR-World uses metrics of decoded predictions as rewards [34], while RLIR introduces inverse-dynamics rewards to encourage action following [39]. These methods enable learning from trajectory feedback, but the quality of the learning signal also depends on *which futures the model compares*. As Figure 1 shows, under the same initialization and update budget, standard GRPO [28] barely improves upon the supervised fine-tuning (SFT) model: insufficient robot motion and pose deviations persist. More candidates therefore do not necessarily yield more informative comparisons.

We attribute this failure to two obstacles in RL for autoregressive world models. The first is *candidate collapse*. Conventional beam search tends to concentrate on nearly identical sequences [31], shrinking within-group reward differences and weakening the learning signal from group-relative advantages, whereas independent sampling does not coordinate candidate quality and scatters into low-likelihood trajectories. The second is *history mismatch*. Once candidates diverge, each branch develops its own robot and object states. When prediction extends beyond the pretraining horizon, the model also relies on bounded memory to sustain long-horizon generation [7, 35, 43]. If policy scoring uses a history that differs from the one that generates a candidate, the optimization no longer corresponds to the trajectory the model actually generates. Effective RL thus requires both high-likelihood, informative differences among candidates and consistent histories for generation and scoring.

How can we construct informative candidate futures and compare them under histories consistent with generation to effectively improve robotic world models?

We propose FUTUREWORLDS (Figure 2), an RL framework built on a multimodal discrete autoregressive world model [30, 33]. Its token probabilities support both candidate expansion and trajectory scoring under visual, action, and language conditioning. To address candidate collapse, FUTUREWORLDS uses diverse beam search during training, explicitly penalizing repetition among candidates in addition to accounting for model likelihood [31]. This produces candidate sets that balance confidence and diversity. To address history mismatch, each branch maintains its own bounded memory: it inherits memory from its parent at branching and independently updates it as generation proceeds. Generation and policy scoring therefore always condition on the same history. As the framework's core training mechanism, MEMSPO (Memory-Conditioned Search-Guided Policy Optimization) converts video trajectory rewards into group-relative advantages [28] and updates the world model using candidates from search. Quality differences among candidates thereby become feedback for improving the generative model itself.

We evaluate FUTUREWORLDS on RT-1 [6], BridgeData V2 [32], and RoboCasa [21]. Under the same initialization and training budget, standard GRPO offers limited improvement over SFT, whereas MEMSPO delivers consistent gains, reducing long-horizon LPIPS by up to 9.75% and 9.98% relative to SFT and GRPO, respectively. Compared with publicly released existing methods on RT-1, the LPIPS reduction of FUTUREWORLDS grows from 2.8% to 14.9% as the prediction horizon increases from 10 to 32 frames. Ablations, optical-flow analysis, and qualitative comparisons further show that these improvements manifest as more accurate motion prediction and more consistent object states. Our main contributions are as follows:

❶ *Problem Diagnosis.* We identify candidate collapse and history mismatch as two key obstacles in RL for autoregressive world models, and analyze how candidate differences determine the learning signal from group-relative advantages.

❷ *Branch-Specific Memory and Search-Guided Policy Optimization.* We propose the FUTUREWORLDS framework and its policy optimization algorithm, MEMSPO: diverse beam search constructs informative candidates, branch-specific memory maintains consistent histories for generation and scoring, and trajectory rewards yield group-relative advantages for updating the world model.

❸ *Experimental Validation.* Across multiple robotic datasets, FUTUREWORLDS substantially outperforms standard GRPO under the same training budget, and its advantage over existing methods grows with the prediction horizon.

2. Method

Figure 2 presents the FUTUREWORLDS framework. A multimodal autoregressive world model first learns conditional video prediction through supervised learning. During reinforcement learning, diverse beam search constructs multiple candidate futures under the same visual, action, and language conditions. Each candidate branch inherits and independently maintains bounded memory from its parent to support long-horizon rollouts. MEMSPO converts decoded video trajectory rewards into group-relative advantages and updates the model under exactly the same branch histories as generation. Ground-truth future frames enter reward computation only and never enter any candidate memory.

2.1. Preliminaries

Robotic world models predict how scenes evolve under action sequences. We use a GPT-style autoregressive model to represent the conditional distribution of future trajectories given the same initial visual conditions, action sequence, and optional language instruction. Let $\mathbf{z}_{1:H}$ denote the discrete

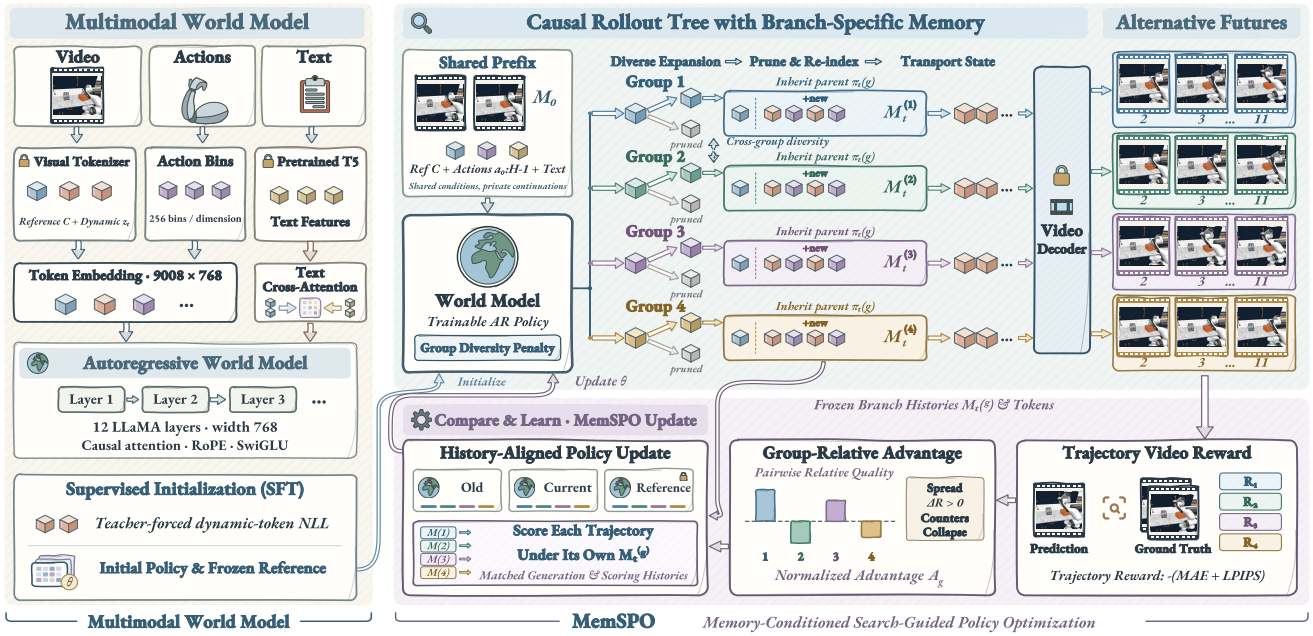


Figure 2. ***Overview of* FUTUREWORLDS.** Left: a multimodal world model with a frozen conditional tokenizer and an autoregressive backbone. Middle: during reinforcement learning, diverse beam search constructs candidate futures, and each branch inherits and independently updates bounded memory from its parent. Right: MEMSPO evaluates policy probabilities under the same branch histories as generation and updates the model using group-relative advantages.

visual sequence for H future frames, with N dynamic tokens per frame. Given initial visual conditions $\mathbf{c}_v$, actions $\mathbf{a}_{0:H-1}$, and a language instruction ℓ, the world model factorizes the distribution as

$$\pi_\theta(\mathbf{z}_{1:H} \mid \mathbf{c}_v, \mathbf{a}_{0:H-1}, \ell) = \prod_{t=1}^{H} \prod_{j=1}^{N} \pi_\theta(z_{t,j} \mid \mathbf{c}_v, \mathbf{z}_{<t}, \mathbf{z}_{t,<j}, \mathbf{a}_{<t}, \ell), \tag{1}$$

Here, $\mathbf{z}_{<t}$ contains tokens from previous frames, $\mathbf{z}_{t,<j}$ is the within-frame prefix, and $\mathbf{a}_{<t} = \mathbf{a}_{0:t-1}$ contains actions before frame t, with a_{t-1} directly determining the transition to that frame. This factorization supports both token-by-token candidate expansion and log-likelihood scoring of complete trajectories.

Following iVideoGPT's conditional video compression design [33], we use more tokens to encode the initial scene and fewer dynamic tokens to encode subsequent changes, reducing repeated encoding of static backgrounds. Context and conditional dynamics encoders produce discrete tokens through Finite Scalar Quantization (FSQ) [20], and a visual decoder reconstructs images from these tokens:

$$C = Q_c\big(E_\phi^c(o_{\text{ref}})\big), \quad \mathbf{z}_t = Q_d\big(E_\phi^d(o_t \mid o_{\text{ref}})\big), \quad \tilde{o}_t = D_\phi(C, \mathbf{z}_t). \tag{2}$$

Here, o_{ref} is the reference frame, and the initial visual conditions $\mathbf{c}_v = (C, \mathbf{z}_{\text{anc}})$ comprise reference-frame context tokens and initial-state tokens. The tokenizer parameters ϕ remain frozen throughout world-model SFT and RL; Appendix C.1 provides the training objective and adaptation details.

Actions undergo per-dimension normalization and discretization into tokens $\tilde{\mathbf{a}}_t$, which interleave with visual tokens in temporal order to form a causal sequence. A frozen T5 encoder [23] supplies language features through cross-attention. Supervised fine-tuning (SFT) conditions on ground-truth histories and applies negative log-likelihood only to target dynamic tokens:

$$\mathcal{L}_{\text{SFT}}(\theta) = -\mathbb{E}_{\mathcal{D}}\left[\frac{1}{HN}\sum_{t=1}^{H}\sum_{j=1}^{N} \log \pi_\theta(z_{t,j} \mid \mathbf{c}_v, \mathbf{z}_{<t}, \mathbf{z}_{t,<j}, \mathbf{a}_{<t}, \ell)\right]. \tag{3}$$

SFT jointly updates the autoregressive backbone, action embeddings, and text cross-attention modules. The resulting model serves as the initial RL policy π_{old}, and a frozen copy serves as the reference policy π_{ref}.

2.2. Branch-Specific Bounded Memory

In multi-candidate rollouts, trajectories develop their own robot poses and object states once they diverge. When rollouts extend beyond the pretraining horizon, the model must maintain history within a bounded context budget B. We

therefore maintain bounded memory $M_t^{(i)}$ for each active branch i. It contains the initial conditions $\mathbf{c}_v$ and pairs of dynamic tokens generated by that branch and corresponding actions from the most recent W frames:

$$M_t^{(i)} = \left[\mathbf{c}_v, \tilde{\mathbf{a}}_0, \left\{\left(\hat{\mathbf{z}}_s^{(i)}, \tilde{\mathbf{a}}_s\right)\right\}_{s\in\mathcal{I}_t}\right], \quad \mathcal{I}_t = \{\max(1, t-W), \ldots, t-1\}, \tag{4}$$

The rolling window length W satisfies $|M_t^{(i)}| \le B$.

Memory inheritance and updates along the search tree. Memory evolves with the search tree rather than remaining a static cache. Suppose branch i, selected at the end of frame t, descends from parent branch $p_t(i)$ at the previous frame. After generating frame tokens $\hat{\mathbf{z}}_t^{(i)}$, the branch updates its memory recursively:

$$M_{t+1}^{(i)} = \mathrm{Trunc}_W\big(M_t^{(p_t(i))} \oplus \big(\hat{\mathbf{z}}_t^{(i)}, \tilde{\mathbf{a}}_t\big)\big), \tag{5}$$

Here, $\oplus$ appends the current predicted frame tokens and action. When the rolling window is full, Trunc_W removes the earliest predicted frame while always retaining the initial conditions $\mathbf{c}_v$. Child branches thus inherit the shared prefix history at branching, while subsequent histories accumulate independently within each branch without interfering with one another.

History alignment between generation and scoring. If all branches share one memory, or scoring substitutes ground-truth histories, policy evaluation conditions differ from those that generate the trajectory, causing *history mismatch*. To eliminate this mismatch, we save each candidate's actual memory trajectory $\{M_t^{(i)}\}_{t=1}^H$ during generation and strictly reuse the same conditions for policy scoring:

$$\log \pi_\theta(\hat{\mathbf{z}}_{1:H}^{(i)} \mid \mathbf{c}_v, \mathbf{a}_{0:H-1}, \ell) = \sum_{t=1}^{H}\sum_{j=1}^{N} \log \pi_\theta(\hat{z}_{t,j}^{(i)} \mid M_t^{(i)}, \hat{\mathbf{z}}_{t,<j}^{(i)}, \ell). \tag{6}$$

The old policy π_{old}, current policy π_θ, and reference policy π_{ref} all evaluate probabilities under the same generation condition $u_{t,j}^{(i)} = (M_t^{(i)}, \hat{\mathbf{z}}_{t,<j}^{(i)}, \ell)$, ensuring that optimization targets the trajectories the model actually generates.

2.3. Memory-Conditioned Search-Guided Policy Optimization (MemSPO)

With history-aligned trajectory scoring, the next step converts quality differences among candidate futures into model updates. We first define the group-relative update objective, then analyze its gradient to explain why candidate collapse eliminates the update signal, and finally describe candidate construction through diverse beam search.

Trajectory rewards and group-relative updates. The frozen visual decoder converts G candidate token sequences into videos $\{\hat{o}_{1:H}^{(i)}\}_{i=1}^G$. Following RLVR-World [34], the base trajectory reward combines pixel error and perceptual distance [42]:

$$R_i = -\frac{1}{H}\sum_{t=1}^{H}\left[\mathrm{MAE}(\hat{o}_t^{(i)}, o_t) + \mathrm{LPIPS}(\hat{o}_t^{(i)}, o_t)\right]. \tag{7}$$

For the G candidates under the same input, we normalize rewards into group-relative advantages and broadcast each advantage to all dynamic tokens in its trajectory:

$$A_i = \frac{R_i - \bar{R}}{\sigma_R + \varepsilon}, \qquad \bar{R} = \frac{1}{G}\sum_{i=1}^{G} R_i, \tag{8}$$

Here, $\sigma_R = \mathrm{Std}(R_1, \ldots, R_G)$ is the within-group reward standard deviation, and $\varepsilon > 0$ is a numerical stabilizer. Under history-aligned conditions, define the token probability ratio $\rho_{t,j}^{(i)}(\theta) = \pi_\theta(\hat{z}_{t,j}^{(i)} \mid u_{t,j}^{(i)})/\pi_{\mathrm{old}}(\hat{z}_{t,j}^{(i)} \mid u_{t,j}^{(i)})$. MemSPO maximizes the group-relative surrogate with reference-policy regularization [28]:

$$\mathcal{J}_{\mathrm{MemSPO}}(\theta) = \frac{1}{GHN}\sum_{i=1}^{G}\sum_{t=1}^{H}\sum_{j=1}^{N} \left[\rho_{t,j}^{(i)}(\theta)A_i - \beta\kappa_{t,j}^{(i)}(\theta)\right], \tag{9}$$

Here, $\kappa = q - \log q - 1$, with $q = \pi_{\mathrm{ref}}/\pi_\theta$, constrains deviation from the reference model. Each iteration constructs a fresh batch of candidates through diverse beam search and performs a single parameter update from θ_{old}, where $\rho_{t,j}^{(i)}(\theta_{\mathrm{old}}) = 1$ and the reward term yields an advantage-weighted log-likelihood gradient.

Vanishing gradients under candidate collapse. Let $\mathbf{s}_i = \frac{1}{HN}\sum_{t,j}\nabla_\theta \log \pi_\theta(\hat{z}_{t,j}^{(i)} \mid u_{t,j}^{(i)})\big|_{\theta=\theta_{\mathrm{old}}}$ denote candidate i's mean token score gradient. Its within-group dispersion is $V_s = \frac{1}{G}\sum_i \|\mathbf{s}_i - \bar{\mathbf{s}}\|_2^2$, where $\bar{\mathbf{s}} = \frac{1}{G}\sum_i \mathbf{s}_i$. Under history alignment between generation and scoring, the reward gradient $\mathbf{g}_R$ of Eq. (9) at $\theta = \theta_{\mathrm{old}}$ admits an equivalent pairwise-difference form and a Cauchy–Schwarz bound (Appendix B):

$$\mathbf{g}_R = \frac{\sum_{i<j}(R_i - R_j)(\mathbf{s}_i - \mathbf{s}_j)}{G^2(\sigma_R + \varepsilon)}, \qquad \|\mathbf{g}_R\|_2 \le \frac{\sigma_R}{\sigma_R + \varepsilon}\sqrt{V_s}. \tag{10}$$

Equation (10) reveals two ways in which candidate collapse weakens learning. First, although advantage normalization removes the absolute reward scale, similar candidate sequences yield similar score gradients ($V_s \to 0$), and $\sqrt{V_s}$ directly drives the update norm toward zero. Second, when candidate rewards barely differ ($\sigma_R \ll \varepsilon$), the factor $\sigma_R/(\sigma_R + \varepsilon)$ also drives the gradient toward zero. Conversely, unconstrained independent sampling enlarges V_s

only by scattering into low-likelihood sequences. Effective group-relative updates therefore require separating candidates within high-probability regions.

Candidate construction through diverse beam search.

To avoid candidate collapse while maintaining high model confidence for individual candidates, we use diverse beam search in RL rollouts [31]. We divide search into G groups, each maintaining b beams. For prefix y and next candidate token v, group g uses the expansion score

$$S_g(y \oplus v) = S_g(y) + \log \pi_{\text{old}}(v \mid u(y)) - \lambda_{\text{div}} d_g(v), \tag{11}$$

Here, $u(y)$ includes the branch's current memory $M_t^{(i)}$ and multimodal conditions, and $d_g(v)$ is the normalized frequency of token v among selections by earlier groups at the same position. Each group retains its b highest-scoring branches and returns one best trajectory at the end of the rollout, yielding G candidate videos that balance likelihood and diversity under the same input. The diversity penalty $-\lambda_{\text{div}} d_g(v)$ affects candidate selection only. Policy scoring and the update in Eq. (9) always use unmodified model probabilities π_θ, maintaining a consistent optimization objective.

3. Experiments

Our experiments address four research questions aligned with our method: ❶ ***RQ1*** How does FUTUREWORLDS perform against existing world models across robotic datasets and prediction horizons? ❷ ***RQ2*** Do diverse beam search and MEMSPO overcome candidate collapse and low-likelihood sampling to outperform standard GRPO under the same post-training budget? ❸ ***RQ3*** Do trajectory-level RL gains translate into more accurate robot motion and more consistent object states? ❹ ***RQ4*** How much does branch-specific bounded memory contribute to long-horizon rollouts beyond the training horizon?

3.1. Experimental Setup

Baselines. We compare against two baseline groups: ✦ robotic world models: FitVid [2], iVideoGPT [33], Vid2World† [17], DriftWorld† [19], and RLVR-World† [34]; and ✦ action-adapted video foundation models: Wan2.2-TI2V-5B [29], Cosmos-Predict2.5-2B [22], and DreamDojo-2B [11]. Methods marked with † use publicly released official checkpoints on RT-1 and are re-evaluated under the same inputs and scoring protocol as our model; evaluation details appear in Appendix C.2.

Datasets and Evaluation. We evaluate action-conditioned video prediction on RT-1 [6], BridgeV2 [32], and RoboCasa [21], covering real and simulated robotic manipulation. Table 1 reports 32-frame PSNR, SSIM, and LPIPS [42] on 128 fixed held-out trajectories per dataset. All methods share identical initial observations, recorded action sequences, and target frames; predicted videos are aligned to the reference resolution and scored by a unified evaluation pipeline, while ground-truth future frames never enter any model's generation history. Metric definitions and multi-horizon (10/15/20/32-frame) evaluation protocols appear in Appendix C.2.

Implementation Details. We train FUTUREWORLDS in PyTorch on 16 NVIDIA H20 GPUs across two servers. Starting from the 50k-update SFT checkpoint, MEMSPO uses AdamW with learning rate 10^{-6} while keeping the visual tokenizer and text encoder frozen. To prevent candidate collapse, RL rollouts use 4×2 diverse beam search ($\lambda_{\text{div}} = 0.1$) and retain one best trajectory per group for group-relative advantage computation and single-step parameter updates. Each branch maintains branch-specific bounded memory consisting of initial conditions and the most recent $W = 6$ frames, strictly reusing identical histories for generation and policy scoring. Main evaluations use beam width four with branch-specific bounded memory. Full hyperparameter settings appear in Appendix C.1.

3.2. Video Generation Quality (RQ1)

As Table 1 shows, FUTUREWORLDS achieves the best PSNR, SSIM, and LPIPS across all three datasets. Relative to the strongest external baseline on each dataset, FUTUREWORLDS reduces 32-frame LPIPS by 14.87%, 20.84%, and 9.12% and improves PSNR by 1.26 dB, 1.71 dB, and 1.80 dB on RT-1, BridgeV2, and RoboCasa, respectively. Compared with its own 50k-update SFT initialization, MEMSPO post-training further lowers LPIPS by 10.57%, 2.69%, and 6.99% across the three benchmarks.

Under the unified RT-1 evaluation against official released checkpoints (Appendix Table 3), as the prediction horizon extends from 10, 15, and 20 frames to 32 frames, FUTUREWORLDS lowers LPIPS from 0.1283/0.1412/0.1571/0.1822 (RLVR-World†) to 0.1247/0.1334/0.1406/0.1551. The relative LPIPS reduction expands monotonically from 2.8% to 14.9%, demonstrating how branch-specific bounded memory and history-aligned optimization curb long-horizon error accumulation. Appendix Figure 7 corroborates this trend at frame 32: whereas baselines exhibit gripper pose drift and object blurring or deformation late in the rollout, FUTUREWORLDS preserves accurate arm trajectories and consistent object states aligned with ground truth.

3.3. Candidate Construction and Post-Training Analysis (RQ2)

Matched-budget post-training comparison. Table 2 compares independent sampling (GRPO), ordinary beam search, and diverse beam search (MEMSPO) across 10, 20, and 32 frames under the same 50k-update SFT initialization, 200-

Table 1. Quantitative comparison of 32-frame action-conditioned video prediction (128 trajectories per dataset). Metrics: PSNR↑, SSIM↑, and LPIPS↓. 1st, 2nd, and 3rd mark the top three, excluding SFT, the 50k-update checkpoint MEMSPO starts from (identical evaluation); † evaluated using official released checkpoints on RT-1.

Method	RT-1			BridgeV2			RoboCasa		
	PSNR	SSIM	LPIPS	PSNR	SSIM	LPIPS	PSNR	SSIM	LPIPS
Robotic World Models									
FitVid	18.71	0.5916	0.4750	17.09	0.6072	0.5235	13.50	0.5422	0.5655
iVideoGPT	15.96	0.5571	0.3833	20.06	0.7791	0.1899	16.67	0.7187	0.2366
Vid2World†	19.67	0.6997	0.2573	18.92	0.7110	0.1880	15.40	0.7000	0.2287
DriftWorld†	20.51	0.7407	0.2597	20.32	0.7429	0.2379	16.50	0.7200	0.2891
RLVR-World†	21.21	0.7783	0.1822	19.06	0.7210	0.1780	15.80	0.7100	0.2017
Adapted Video Foundation Models									
Wan2.2-TI2V-5B	18.42	0.7058	0.2478	16.35	0.6889	0.2568	14.37	0.6906	0.2711
Cosmos-Predict2.5-2B	17.77	0.6765	0.2889	18.98	0.7799	0.1870	14.30	0.6802	0.2916
DreamDojo-2B	18.47	0.7152	0.2660	16.95	0.7461	0.2563	13.92	0.6349	0.3278
SFT initialization (ours)	21.37	0.7860	0.1734	21.74	0.7942	0.1448	17.50	0.7519	0.1971
FutureWorlds (ours)	**22.47**	**0.8026**	**0.1551**	**22.03**	**0.7968**	**0.1409**	**18.47**	**0.7659**	**0.1833**
Gain over best baseline	+5.96%	+3.12%	+14.87%	+8.42%	+2.17%	+20.84%	+10.80%	+6.38%	+9.12%
Gain over SFT	+5.17%	+2.11%	+10.57%	+1.33%	+0.33%	+2.69%	+5.58%	+1.87%	+6.99%

Table 2. Post-training comparison under a matched 200-update budget (128 trajectories per dataset). Metrics: PSNR↑, SSIM↑, and LPIPS↓. All RL variants start from the same 50k-update SFT checkpoint and share identical reward, memory, and evaluation settings. Gain rows report relative improvements of MEMSPO over standard GRPO and LPIPS reductions over ordinary beam search.

Method (updates)	10 frames			20 frames			32 frames		
	PSNR	SSIM	LPIPS	PSNR	SSIM	LPIPS	PSNR	SSIM	LPIPS
RT-1									
SFT (50k)	23.76	0.8298	0.1308	22.49	0.8075	0.1527	21.37	0.7860	0.1734
GRPO (200)	23.97	0.8319	0.1289	22.60	0.8089	0.1517	21.39	0.7858	0.1739
Ordinary beam (200)	24.03	0.8329	0.1279	22.85	0.8124	0.1479	21.70	0.7908	0.1690
MemSPO (200)	**24.33**	**0.8357**	**0.1240**	**23.42**	**0.8192**	**0.1397**	**22.45**	**0.8016**	**0.1565**
Gain over GRPO	+1.49%	+0.45%	+3.80%	+3.61%	+1.28%	+7.92%	+4.97%	+2.01%	+9.98%
LPIPS reduction vs. Ordinary beam		3.04%			5.54%			7.37%	
BridgeV2									
SFT (50k)	24.06	0.8300	0.1095	22.74	0.8085	0.1270	21.74	0.7942	0.1448
GRPO (200)	24.11	0.8300	0.1090	22.78	0.8086	0.1266	21.79	0.7944	0.1442
Ordinary beam (200)	24.08	0.8298	0.1095	22.86	0.8091	0.1259	21.89	0.7956	0.1424
MemSPO (200)	**24.31**	**0.8320**	**0.1073**	**22.99**	**0.8108**	**0.1244**	**21.95**	**0.7960**	**0.1421**
Gain over GRPO	+0.83%	+0.23%	+1.56%	+0.92%	+0.27%	+1.74%	+0.73%	+0.20%	+1.46%
LPIPS reduction vs. Ordinary beam		1.99%			1.18%			0.20%	
RoboCasa									
SFT (50k)	19.50	0.7957	0.1513	18.22	0.7689	0.1787	17.50	0.7519	0.1971
GRPO (200)	19.68	0.7965	0.1514	18.64	0.7729	0.1758	18.03	0.7581	0.1919
Ordinary beam (200)	19.69	**0.7985**	0.1490	18.57	**0.7746**	0.1745	17.98	**0.7607**	0.1906
MemSPO (200)	**19.92**	0.7984	**0.1476**	**18.88**	0.7743	**0.1718**	**18.29**	0.7600	**0.1877**
Gain over GRPO	+1.21%	+0.24%	+2.47%	+1.31%	+0.19%	+2.28%	+1.45%	+0.25%	+2.17%
LPIPS reduction vs. Ordinary beam		0.94%			1.53%			1.51%	

update RL budget, and evaluation settings. On the real-robot datasets, standard GRPO barely improves over SFT (e.g., 32-frame LPIPS of 0.1739 versus 0.1734 on RT-1), and ordinary beam search yields moderate gains (0.1690). In contrast, MEMSPO achieves the best PSNR and LPIPS at every horizon across all three datasets, lowering 32-frame LPIPS by 9.98%, 1.46%, and 2.17% relative to GRPO and by 7.37%, 0.20%, and 1.51% relative to ordinary beam search on RT-1, BridgeV2, and RoboCasa, respectively. Only 200 MEMSPO updates trained on 10-frame rollouts thus consistently im-

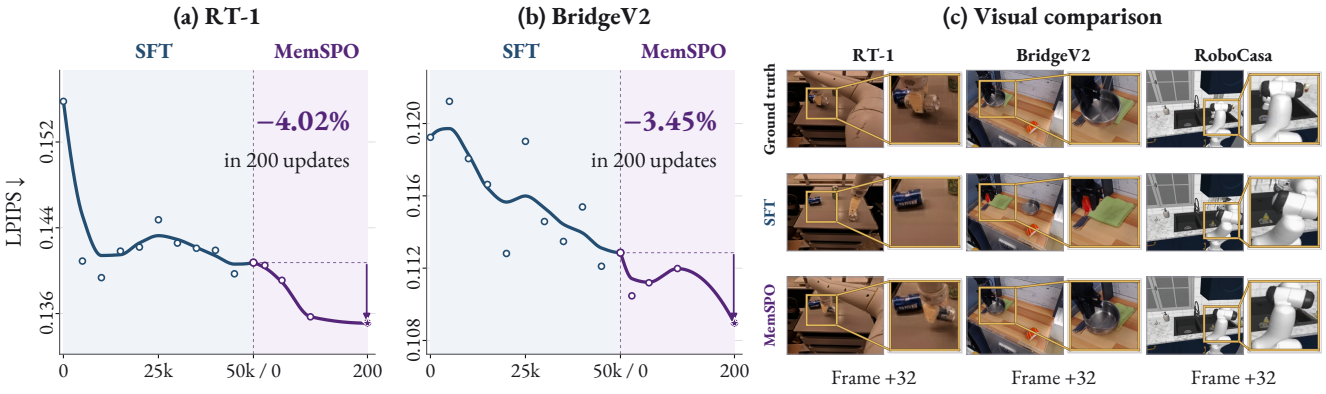


Figure 3. ***Post-training gains and qualitative improvements.*** (a,b) 10-frame LPIPS during SFT and MemSPO training. (c) Frame-32 predictions with matched detail views on three datasets.

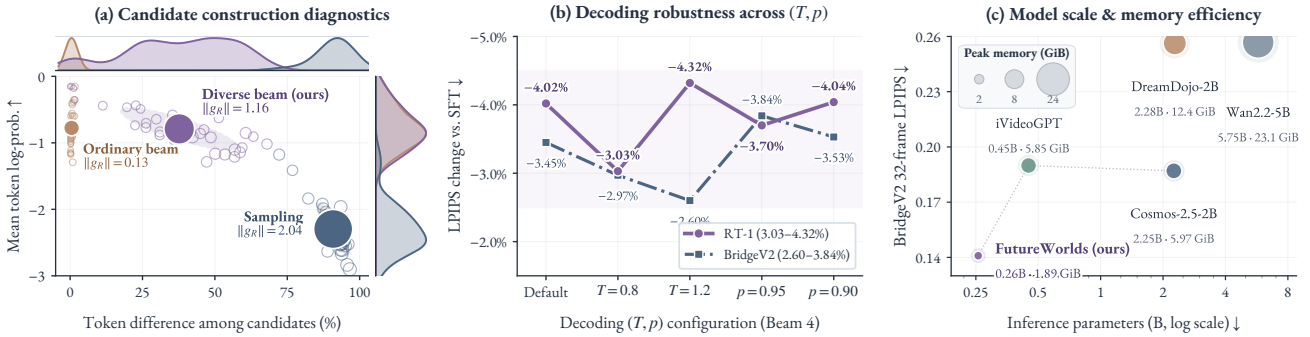


Figure 4. ***Candidate construction diagnostics, decoding robustness, and resource efficiency.*** (a) Candidate construction at the frozen RT-1 SFT checkpoint across 32 fixed training inputs (10-frame rollouts, 4 retained candidates per input); large solid circles mark means, hollow circles and dashed ellipses show per-input distributions, top/right curves show marginal kernel densities, and marker area is proportional to the local reward-gradient norm $\|\mathbf{g}_R\|_2$. (b) Relative 10-frame LPIPS reduction of MemSPO over SFT across decoding temperature T and top-p configurations. (c) Model scale, peak GPU memory on one H20, and 32-frame LPIPS on BridgeV2.

prove 32-frame generation well beyond the training horizon.

Mechanism diagnostics of candidate collapse and gradient signals. To verify the mechanism in Eq. (10), Figure 4(a) directly measures pairwise token difference, mean token log-probability $\log \pi_{\text{old}}$, and the local reward-gradient norm $\|\mathbf{g}_R\|_2$ across the three candidate construction strategies on the frozen RT-1 SFT checkpoint. The joint and marginal distributions separate three distinct regimes. *Ordinary beam search* maintains high token log-likelihood ($\log \pi_{\text{old}} = -0.78$) but collapses to a 0.50% pairwise token difference, which shrinks score-gradient dispersion ($\sqrt{V_s} = 0.23$) and compresses the reward-gradient norm to $\|\mathbf{g}_R\|_2 = 0.13$, directly confirming the bound in Eq. (10). Conversely, *independent sampling (GRPO)* produces high token diversity (90.85%) and a large gradient norm (2.04), but its mean token log-likelihood drops sharply to -2.30, concentrating comparisons on low-confidence trajectories. *Diverse beam search* preserves high token log-likelihood (-0.79, matching ordinary beam search) while raising token difference to 37.70%, expanding $\sqrt{V_s}$ to 2.04 (8.9×) and restoring $\|\mathbf{g}_R\|_2$ to 1.16 (9.1×) to supply high-confidence, informative updates.

Training dynamics, decoding robustness, and resource footprint. As Figure 3(a,b) shows, 200 MemSPO updates steadily reduce 10-frame LPIPS by 4.02% and 3.45% relative to SFT on RT-1 and BridgeV2 monitoring clips and substantially improve frame-32 object states and robot arm poses (Figure 3c). Across five decoding temperatures and top-p thresholds, these relative LPIPS reductions remain stable at 3.03–4.32% on RT-1 and 2.60–3.84% on BridgeV2 (Figure 4b; Appendix Table 4). Furthermore, owing to compressive conditional tokenization and a compact autoregressive backbone, FutureWorlds requires only 0.26B inference parameters and 1.89 GiB peak GPU memory while outperforming baseline world models and adapted video foundation models ranging from 0.45B to 5.75B parameters and 5.85 to 23.1 GiB peak memory on 32-frame BridgeV2 generation (Figure 4c).

3.4. Motion and Dynamic Prediction (RQ3)

Using WorldArena's optical-flow and feature diagnostics [27], we evaluate 128 trajectories per dataset against

(a) Motion prediction quality

32 frames · 128 trajectories per dataset

Method	RT-1			BridgeV2			RoboCasa		
	Flow ↓	Gap ↓	DINO ↑	Flow ↓	Gap ↓	DINO ↑	Flow ↓	Gap ↓	DINO ↑
SFT	1.29	0.74	0.94	0.90	0.33	0.94	1.28	0.81	0.92
Cosmos-Predict2.5-2B	2.03	0.89	0.81	1.72	0.53	0.88	1.99	1.04	0.77
DreamDojo-2B	2.34	0.78	0.85	1.92	0.59	0.83	2.94	1.48	0.85
FutureWorlds (ours)	**1.13**	**0.43**	**0.96**	**0.86**	**0.30**	**0.94**	**1.07**	**0.53**	**0.92**

(b) Optical flow

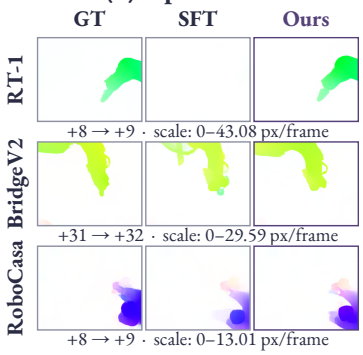


Figure 5. ***Motion quality and optical-flow comparisons.*** (a) Full-cohort results at 32 frames for SFT, Cosmos, DreamDojo, and FutureWorlds. (b) Ground-truth, SFT, and FutureWorlds flow examples with a shared color scale within each dataset.

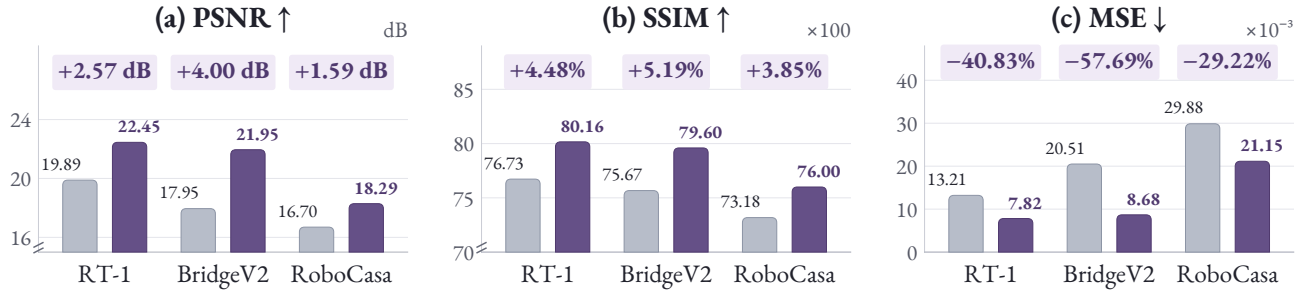


Figure 6. ***Branch-specific bounded memory ablation at 32 frames.*** w/o Rolling History and Branch-Specific Bounded Memory (Ours). Callouts report gains of bounded memory over removing rolling history.

SFT, Cosmos, and DreamDojo (Figure 5a). Flow error measures adjacent-frame motion-vector disagreement, magnitude gap measures motion-strength deviation, and DINO similarity measures frame-wise semantic consistency (Appendix C.2). FUTUREWORLDS achieves the lowest flow error and magnitude gap and matches or improves the highest DINO consistency, reducing flow error by 12.40%, 4.57%, and 16.68% relative to SFT on RT-1, BridgeV2, and RoboCasa, respectively. Simultaneous reductions in flow vector error and magnitude gap confirm that MEMSPO improves both motion direction and amplitude rather than merely amplifying movement, as visually corroborated by the shared-scale optical-flow comparisons in Figure 5(b).

3.5. Branch-Specific Bounded Memory (RQ4)

To evaluate how branch-specific bounded memory supports rollouts well beyond the 10-frame training window, we compare retaining the rolling self-generated history ($W = 6$ recent frames alongside initial conditions) against retaining only the latest single-frame state (*w/o Rolling History*). As Figure 6 shows, branch-specific bounded memory substantially improves 32-frame PSNR, SSIM, and MSE across all three datasets, boosting PSNR by 2.57 dB, 4.00 dB, and 1.59 dB and reducing MSE by 40.83%, 57.69%, and 29.22% on RT-1, BridgeV2, and RoboCasa, respectively. Maintaining branch-specific rolling history is therefore essential for preserving scene geometry and object states over long-horizon rollouts.

4. Conclusion and Future Work

We present FUTUREWORLDS, which reinforces multimodal autoregressive robotic world models by jointly resolving candidate collapse and history mismatch. Our gradient analysis and empirical probe show that conventional beam search suffers from vanishing group-relative gradients under candidate collapse, whereas independent sampling scatters into low-likelihood trajectories. To overcome both obstacles, MEMSPO combines diverse beam search with branch-specific bounded memory inherited along the search tree, updating the model with group-relative advantages under identical generation and scoring histories. Experiments across RT-1, BridgeV2, and RoboCasa demonstrate that FUTUREWORLDS improves visual quality, motion accuracy, and object-state consistency within a compact memory footprint, with gains over existing methods expanding as the prediction horizon grows. Coordinating candidate construction with branch-level history maintenance thus enables world models to learn effectively from their own rollouts. Future work will extend FUTUREWORLDS to closed-loop robot control.

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FutureWorlds: Branch-Specific Memory for Long-Horizon Robotic World Models

Supplementary Material

A. Related Work

A.1. Robotic World Models

Robotic world models predict action consequences for planning and policy learning. Early work uses action-conditioned video prediction for visual control [9, 10]; PlaNet [14] and Dreamer [15] learn latent dynamics. UniSim [38] combines language and low-level controls for visual simulation, while Vid2World [17] and DriftWorld [19] adapt diffusion and drifting generation for interactive robotic prediction. VideoGPT [37] and iVideoGPT [33] develop discrete autoregressive world models, with iVideoGPT integrating observations, actions, and rewards through compressive conditional tokenization. FUTUREWORLDS builds on discrete autoregression to construct and compare candidate futures under shared visual, action, and language conditions.

A.2. Memory for Long-Horizon World Modeling

Transformer-XL [7] propagates segment-level history, while StreamingLLM [36] retains initial attention sinks and recent context. StreamingT2V [16] combines short- and long-term memory; FramePack [41] packs historical frames into a fixed context budget. Wu et al. [35] introduce geometry-grounded memory for scene revisits, and Ctrl-World [13] and Mem-World [43] retrieve robotic observations through pose conditioning and spatiotemporal surfel indexing. Unlike models that maintain a single linear history, FUTUREWORLDS maintains branch-specific bounded memory that is inherited from parent branches along the search tree and updated independently during self-generated rollouts, strictly reusing identical histories for generation and policy scoring to prevent history mismatch.

A.3. Search-Guided Reinforcement Learning for World Models

MIXER [24] and SCST [25] optimize sequence metrics through policy gradients, and DDPO [5] extends reward optimization to diffusion models. In world modeling, RLVR-World [34] rewards decoded prediction quality, RLIR [39] evaluates action following through inverse dynamics, and PersistWorld [3] explores RL post-training to stabilize multi-step rollouts. However, the quality of the learning signal depends on which candidate futures the model compares: conventional beam search collapses onto nearly identical sequences, whereas independent sampling scatters into low-confidence trajectories. Diverse Beam Search [31] reduces redundancy through cross-group penalties, Expert Iteration [1] alternates search and policy learning, BGRPO [18] combines beam-generated candidates with rank-aware rewards for algebraic tasks, and QWM [8] searches predicted futures to select robotic actions. Rather than searching over actions, FUTUREWORLDS constructs high-likelihood, diverse candidate futures under a fixed action sequence, and MEMSPO converts their trajectory rewards into group-relative advantages [28] under matched branch histories to update the world model.

B. Reward-Gradient Analysis

This section derives the pairwise gradient identity and the norm bound under candidate collapse in Eq. (10), showing how differences within the candidate set determine the local update signal of MEMSPO.

B.1. Pairwise Decomposition of the Reward Gradient

For a fixed input, consider $G \geq 2$ candidate trajectories, each containing $L = HN$ dynamic tokens. Flatten the token indices into $k \in \{1, \dots, L\}$, and let $z_k^{(i)}$ and $u_k^{(i)}$ denote candidate i's token and history-aligned condition, respectively. The condition includes branch memory $M_t^{(i)}$, the within-frame prefix, and language conditioning. Candidate tokens, history conditions, trajectory rewards, and normalization statistics remain fixed during differentiation.

Following Eq. (8), let $\sigma_R = \mathrm{Std}(R_1, \dots, R_G)$ be the sample standard deviation. Define the group-relative advantages and mean token score gradients as

$$A_i = \frac{R_i - \bar{R}}{\sigma_R + \varepsilon}, \qquad \bar{R} = \frac{1}{G}\sum_{i=1}^{G} R_i,$$
$$\mathbf{s}_i = \frac{1}{L}\sum_{k=1}^{L} \nabla_\theta \log \pi_\theta(z_k^{(i)} \mid u_k^{(i)})\Big|_{\theta=\theta_{\mathrm{old}}}, \tag{B.1}$$

Here, $\varepsilon > 0$ is a numerical stabilizer. The reward component of Eq. (9) is

$$\mathcal{J}_R(\theta) = \frac{1}{GL}\sum_{i=1}^{G}\sum_{k=1}^{L} \rho_k^{(i)}(\theta) A_i,$$
$$\rho_k^{(i)}(\theta) = \frac{\pi_\theta(z_k^{(i)} \mid u_k^{(i)})}{\pi_{\mathrm{old}}(z_k^{(i)} \mid u_k^{(i)})}. \tag{B.2}$$

When the numerator and denominator evaluate probabilities under the same generation history $u_k^{(i)}$, $\rho_k^{(i)}(\theta_{\mathrm{old}}) = 1$. Using the log-derivative identity $\nabla_\theta \rho_k^{(i)}(\theta) = \rho_k^{(i)}(\theta) \nabla_\theta \log \pi_\theta(z_k^{(i)} \mid u_k^{(i)})$, substituting into Eq. (B.2),

and evaluating at $\theta = \theta_{\text{old}}$ gives the centered gradient:

$$\begin{aligned}\mathbf{g}_R &:= \nabla_\theta \mathcal{J}_R(\theta)|_{\theta=\theta_{\text{old}}} \\ &= \frac{1}{G}\sum_{i=1}^{G} A_i \mathbf{s}_i \\ &= \frac{1}{G(\sigma_R+\varepsilon)}\sum_{i=1}^{G}(R_i-\bar{R})\mathbf{s}_i.\end{aligned} \tag{B.3}$$

We next express this gradient through pairwise differences. Index symmetry and cancellation of the diagonal terms for $i = j$ give

$$\begin{aligned}&\sum_{i<j}(R_i-R_j)(\mathbf{s}_i-\mathbf{s}_j) \\ &= \frac{1}{2}\sum_{i=1}^{G}\sum_{j=1}^{G}\big(R_i\mathbf{s}_i + R_j\mathbf{s}_j \\ &\quad - R_i\mathbf{s}_j - R_j\mathbf{s}_i\big) \\ &= G\sum_{i=1}^{G} R_i\mathbf{s}_i - \left(\sum_{i=1}^{G} R_i\right)\left(\sum_{i=1}^{G}\mathbf{s}_i\right) \\ &= G\sum_{i=1}^{G}(R_i-\bar{R})\mathbf{s}_i.\end{aligned} \tag{B.4}$$

Substituting Eq. (B.4) into Eq. (B.3) yields the first equality in Eq. (10):

$$\mathbf{g}_R = \frac{\sum_{i<j}(R_i-R_j)(\mathbf{s}_i-\mathbf{s}_j)}{G^2(\sigma_R+\varepsilon)}. \tag{B.5}$$

This identity applies to any candidate set, including trajectories constructed by diverse beam search, and history alignment ensures that each $\mathbf{s}_i$ evaluates the conditional distribution that generates candidate i.

B.2. Magnitude Bound under Candidate Collapse

Define the within-group population variance of rewards and the dispersion of score gradients as

$$\begin{aligned}\bar{\mathbf{s}} &= \frac{1}{G}\sum_{i=1}^{G}\mathbf{s}_i, \\ V_R &= \frac{1}{G}\sum_{i=1}^{G}(R_i-\bar{R})^2, \\ V_s &= \frac{1}{G}\sum_{i=1}^{G}\|\mathbf{s}_i-\bar{\mathbf{s}}\|_2^2.\end{aligned} \tag{B.6}$$

Since $\sum_{i=1}^{G}(R_i-\bar{R}) = 0$, subtracting the mean vector $\bar{\mathbf{s}}$ in Eq. (B.3) leaves the gradient unchanged:

$$\mathbf{g}_R = \frac{1}{G(\sigma_R+\varepsilon)}\sum_{i=1}^{G}(R_i-\bar{R})(\mathbf{s}_i-\bar{\mathbf{s}}). \tag{B.7}$$

The triangle inequality and the Cauchy–Schwarz inequality give

$$\begin{aligned}&\|\mathbf{g}_R\|_2 \\ &\le \frac{1}{G(\sigma_R+\varepsilon)}\sum_{i=1}^{G}|R_i-\bar{R}|\,\|\mathbf{s}_i-\bar{\mathbf{s}}\|_2 \\ &\le \frac{\sqrt{\sum_i(R_i-\bar{R})^2}\sqrt{\sum_i\|\mathbf{s}_i-\bar{\mathbf{s}}\|_2^2}}{G(\sigma_R+\varepsilon)} \\ &= \frac{\sqrt{V_R V_s}}{\sigma_R+\varepsilon}.\end{aligned} \tag{B.8}$$

Since $\sqrt{V_R} = \sqrt{\frac{G-1}{G}}\,\sigma_R \le \sigma_R$, substitution into Eq. (B.8) yields the bound in Eq. (10):

$$\begin{aligned}\|\mathbf{g}_R\|_2 &\le \frac{\sigma_R}{\sigma_R+\varepsilon}\sqrt{\frac{G-1}{G}\,V_s} \\ &\le \frac{\sigma_R}{\sigma_R+\varepsilon}\sqrt{V_s}.\end{aligned} \tag{B.9}$$

The score-gradient dispersion also admits a pairwise-distance expansion:

$$V_s = \frac{1}{G^2}\sum_{i<j}\|\mathbf{s}_i-\mathbf{s}_j\|_2^2. \tag{B.10}$$

Thus, when all candidate score gradients converge ($\|\mathbf{s}_i - \mathbf{s}_j\|_2 \to 0$, equivalently $V_s \to 0$), or all candidate rewards converge ($\sigma_R \to 0$), $\|\mathbf{g}_R\|_2 \to 0$.

C. Implementation and Evaluation Details

C.1. Initialization and Training

Tokenizer. We initialize from publicly released visual-tokenizer and autoregressive world-model checkpoints. Tokenizer training jointly optimizes pixel reconstruction, perceptual, and adversarial losses:

$$\mathcal{L}_{\text{tok}}(\phi) = \lambda_{\text{rec}}\mathcal{L}_1 + \lambda_{\text{perc}}\mathcal{L}_{\text{LPIPS}} + \lambda_{\text{adv}}\mathcal{L}_{\text{GAN}}. \tag{C.1}$$

Here, $\lambda_{\text{rec}} = 1.0$, $\lambda_{\text{perc}} = 1.0$, and $\lambda_{\text{adv}} = 0.1$, and the discriminator uses a hinge loss with gradient penalty. The tokenizer remains frozen throughout world-model SFT and RL.

SFT and RL. Supervised fine-tuning runs for 50,000 updates with global batch size 64, backbone learning rate 10^{-5}, 500 warmup updates, and gradient-norm limit 1.0, supervising dynamic tokens under recorded histories. SFT initializes both the RL policy and the frozen reference policy. During RL, MEMSPO uses ten-frame rollouts, global batch size 16, AdamW with learning rate 10^{-6}, reference coefficient 0.001, and gradient-norm limit 1.0. Diverse beam search uses $G = 4$ groups of $b = 2$ beams with diversity penalty $\lambda_{\text{div}} = 0.1$, retaining the highest-scoring trajectory in each group for reward evaluation and single-step updates under saved generation-time branch histories.

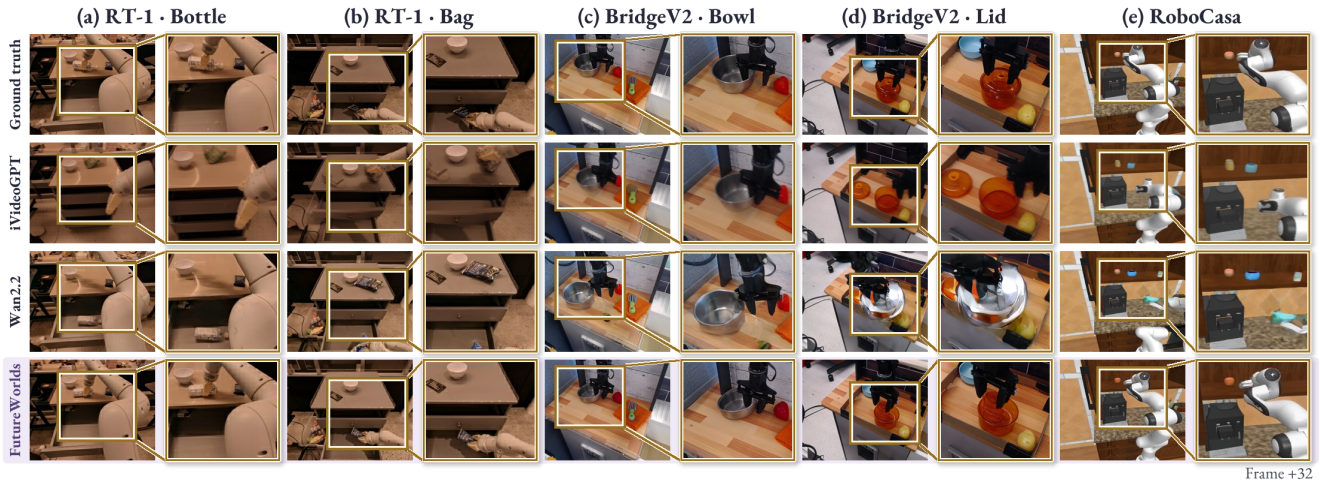


Figure 7. ***Qualitative comparison at frame 32.*** Five representative cases on RT-1, BridgeV2, and RoboCasa comparing two baselines and our method, with full images and matched detail views.

Table 3. **Multi-horizon LPIPS↓ against official RT-1 checkpoints** (128 trajectories, unified evaluation).

Method	10f	15f	20f	32f
Vid2World†	0.1784	0.1958	0.2124	0.2573
DriftWorld†	0.1863	0.2072	0.2240	0.2597
RLVR-World†	0.1283	0.1412	0.1571	0.1822
FutureWorlds (ours)	**0.1247**	**0.1334**	**0.1406**	**0.1551**
Reduction vs. RLVR-World†	2.8%	5.5%	10.5%	14.9%

C.2. Evaluation Protocol and Additional Results

Metrics and multi-horizon evaluation. Table 1 evaluates 128 fixed 32-frame trajectories per dataset. Predictions are aligned to the reference resolution (256×320), and frame-wise scores are averaged over time and trajectories. PSNR is $-10\log_{10}(\mathrm{MSE})$ for RGB in $[0, 1]$; SSIM measures structural similarity; LPIPS-VGG [42] evaluates perceptual distance on inputs in $[-1, 1]$. Recorded actions condition generation, and ground-truth future images never enter candidate histories. Table 3 reports multi-horizon LPIPS on RT-1 against official baseline checkpoints across 10, 15, 20, and 32 predicted frames, and Figure 7 provides qualitative comparisons at frame 32 across RT-1, BridgeV2, and RoboCasa.

Motion diagnostics. Figure 5 evaluates the same 128 held-out trajectories per dataset over 32 frames using WorldArena's RAFT optical-flow and DINO feature components [27]: Flow measures mean Euclidean optical-flow vector error between adjacent frames, Gap measures the absolute difference in mean flow magnitudes, and DINO measures frame-wise feature cosine similarity.

C.3. Decoding Robustness

Table 4 verifies that MEMSPO consistently improves upon SFT across decoding temperatures and top-p thresholds.

Table 4. Relative 10-frame LPIPS reduction (%) of MEMSPO over SFT across decoding settings (32 fixed clips per dataset, Beam 4).

Temperature T	Top-p	RT-1	BridgeV2
1.0	1.00	4.02	3.45
0.8	1.00	3.03	2.97
1.2	1.00	4.32	2.60
1.0	0.95	3.70	3.84
1.0	0.90	4.04	3.53